

Design and Simulation of 2-bit Full Adder.

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Abstract—Adders are fundamental components in digital electronics, forming the basis of arithmetic operations in processors, embedded systems, and digital signal processing units. A 2-bit full adder is a crucial combinational logic circuit that enables the addition of two binary numbers, each consisting of two bits, along with a carry-in input. The resulting output includes a 2-bit sum and a carry-out, which together represent the complete binary addition result. This project involves the comprehensive design, simulation, and layout implementation of a 2-bit full adder using Cadence Virtuoso, an industry-standard EDA tool. The design flow includes behavioral-level schematic design, gate-level and transistor-level implementation, symbol generation, functional and post-layout simulations, and physical layout creation. Additionally, critical verification steps such as Design Rule Check (DRC), Layout Versus Schematic (LVS), and parasitic extraction were carried out to ensure design accuracy and manufacturability. The final chip layout includes integrated I/O pads and is ready for tape-out, highlighting the successful completion of a full custom digital design process. This report serves as a valuable reference for understanding the end-to-end design and verification of a moderately complex digital circuit in VLSI design

Index Terms—2-bit full adder, Cadence Virtuoso, schematic, layout, DRC, LVS, GDSII, transistor-level design, digital IC, CMOS

I. INTRODUCTION

The 2-bit full adder is an essential building block for arithmetic logic units, which are integral to all forms of digital processing. The full adder circuit adds three binary bits and produces a sum and a carry bit. By cascading two 1-bit full adders, we can design a 2-bit full adder capable of computing the sum of two 2-bit numbers along with an optional carry-in. This approach introduces the concept of hierarchical design, which is a foundational principle in digital VLSI design. This report outlines the design and implementation of a 2-bit full adder using the Cadence Virtuoso platform. The workflow begins with the behavioral schematic of the full adder and follows through to the creation of its transistor-level layout, verification of physical and logical accuracy, and finally the generation of a GDSII file suitable for fabrication. By using a structured design methodology, this

project not only implements the required functionality but also ensures manufacturability and design integrity.

II. METHODOLOGY

A. 2 Bit FULL ADDER DESIGN

The Full-Adder (FA) is a critical component extensively utilized in digital systems for performing arithmetic operations such as counting, addition, subtraction, multiplication, and division. It serves as the foundational building block of the Arithmetic Logic Unit (ALU), and advancements in its design directly impact the overall processing speed of the Central Processing Unit (CPU). Among all digital logic operations, addition is particularly vital, and the performance of a digital system heavily relies on the speed and accuracy of its adder circuit. Various logic styles such as pseudo-NMOS and pass-transistor logic can help minimize the transistor count in implementing logic functions. However, these approaches often come at the cost of increased power dissipation. XOR gates, essential for adder design, can be implemented using CMOS inverters and pass-transistor configurations. Additionally, techniques like the dual sleep approach have been explored to enhance power efficiency. In this project, all designs were developed using the **GPDK 90nm process technology**, ensuring a realistic and scalable VLSI implementation. Transistor sizing, defined by the width-to-length (W/L) ratio, significantly influences parameters like sub-threshold leakage current and switching performance. When sleep transistors are used, it becomes necessary to evaluate ground bounce noise and leakage characteristics, as sub-threshold current is proportional to the W/L ratio of the transistor.

B. Boolean Expressions for 2-bit Full Adder

$$S_0 = A_0 \oplus B_0 \oplus C_{in} \quad (1)$$

$$C_1 = (A_0 \cdot B_0) + (C_{in} \cdot (A_0 \oplus B_0)) \quad (2)$$

$$S_1 = A_1 \oplus B_1 \oplus C_1 \quad (3)$$

$$C_{out} = (A_1 \cdot B_1) + (C_1 \cdot (A_1 \oplus B_1)) \quad (4)$$

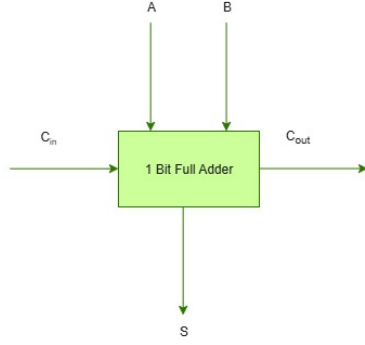


Fig. 1. 1-bit Full Adder Circuit Block

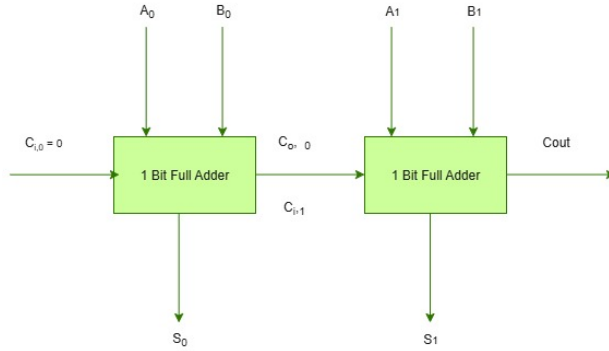


Fig. 2. 2-bit Full Adder Circuit Block

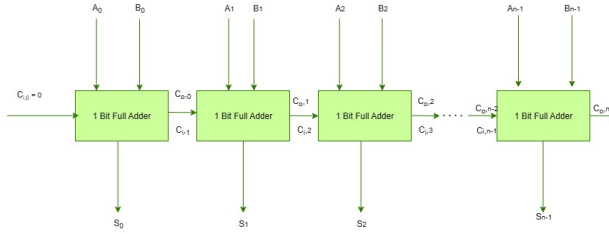


Fig. 3. Combinational Adder for Multiple Bits

C. Truth Table of 2-bit Full Adder

INPUT					OUTPUT		
A ₀	A ₁	B ₀	B ₁	C _{in}	S ₁	S ₀	C _{out}
0	0	0	0	0	0	0	0
0	0	0	0	1	0	1	0
0	0	0	1	0	1	0	0
0	0	0	1	1	1	1	0
0	0	1	0	0	1	0	0
0	0	1	0	1	1	1	0
0	0	1	1	0	1	1	0
0	0	1	1	1	0	0	1
0	1	0	0	0	1	0	0
0	1	0	0	1	1	1	0
0	1	0	1	0	1	0	1
0	1	0	1	1	0	0	1
0	1	1	0	0	0	1	0
0	1	1	0	1	0	0	1
0	1	1	1	0	0	0	1
0	1	1	1	1	0	1	1
1	0	0	0	0	1	0	0
1	0	0	0	1	1	1	0
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1	0	0	1	1	0	0	1
1	0	1	0	0	0	1	0
1	0	1	0	1	0	0	1
1	0	1	1	0	0	0	1
1	0	1	1	1	0	1	1
1	1	0	0	0	0	1	0
1	1	0	0	1	0	0	1
1	1	0	1	0	0	1	1
1	1	0	1	1	0	0	1
1	1	1	0	0	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	0	0	1	1
1	1	1	1	1	1	0	1

D. 2-bit Full Adder Design Description

The 2-bit full adder is a combinational digital circuit designed to add two binary numbers, each 2 bits wide, along with an optional carry-in input (C_{in}). The resulting output includes a 2-bit sum (S_1 and S_0) and a carry-out (C_{out}). The architectural design of the 2-bit full adder is based on the modular composition of two 1-bit full adder units.

STEP 1: INPUT AND OUTPUT DEFINITION

• Inputs:

- A_1, A_0 – First 2-bit binary input
- B_1, B_0 – Second 2-bit binary input
- C_{in} – Initial carry input (from a previous stage or set to 0)

• Outputs:

- S_1, S_0 – 2-bit binary sum output
- C_{out} – Final carry output after the addition

Each full adder module performs a 3-bit addition (two data bits and one carry input), so the design uses two full adders in cascade.

STEP 2: HIERARCHICAL BLOCK DIAGRAM

The design is implemented using two 1-bit full adder blocks:

1) First Full Adder (LSB Adder)

- Inputs: A_0, B_0, C_{in}
- Outputs: S_0 (Sum of LSB), Carry-out (C_1)

2) Second Full Adder (MSB Adder)

- Inputs: A_1, B_1, C_1 (carry propagated from LSB adder)
- Outputs: S_1 (Sum of MSB), C_{out} (final carry-out)

This cascaded architecture supports carry propagation, a fundamental principle in digital arithmetic operations.

STEP 3: LOGICAL GATE CONSTRUCTION

Each 1-bit full adder is internally constructed using the following gates:

- XOR Gates: For sum calculation
- AND and OR Gates: For carry calculation

Logical expressions for each stage:

• First Full Adder:

$$S_0 = A_0 \oplus B_0 \oplus C_{in} \quad (5)$$

$$C_1 = (A_0 \cdot B_0) + (C_{in} \cdot (A_0 \oplus B_0)) \quad (6)$$

• Second Full Adder:

$$S_1 = A_1 \oplus B_1 \oplus C_1 \quad (7)$$

$$C_{out} = (A_1 \cdot B_1) + (C_1 \cdot (A_1 \oplus B_1)) \quad (8)$$

These expressions are mapped to gate-level schematic designs using CMOS logic in Cadence Virtuoso.

STEP 4: IMPLEMENTATION TECHNOLOGY

- Technology Used: GPDK 90nm CMOS
- Transistor sizing (W/L ratio) was optimized to balance speed and power.
- The design was implemented and verified through schematic entry, symbol creation, and simulation.

STEP 5: DESIGN REUSABILITY AND MODULARITY

- The use of hierarchical design with reusable symbols for the 1-bit full adder allows scalable architectures for n-bit adders.
- Modular design also enables easier debugging, layout generation, and design rule checks.

E. State Diagram applicability to a 2-bit Full Adder

A state diagram is typically used to represent the behavior of sequential circuits, which depend not only on the current inputs but also on previous inputs or internal states. These diagrams illustrate transitions between different states based on inputs and are essential in modeling finite state machines (FSMs), memory elements, or control logic.

However, in the case of a 2-bit full adder, a state diagram is not applicable. This is because the 2-bit full adder is a purely combinational circuit, meaning:

- Its outputs (S_1 , S_0 , and C_{out}) depend only on the present values of the inputs (A_1 , A_0 , B_1 , B_0 , and C_{in}).
- There is no memory element or feedback, and it does not retain any state from previous operations.
- As such, it does not exhibit any form of state transition over time.

F. SCHEMATIC LAYOUT DESIGN

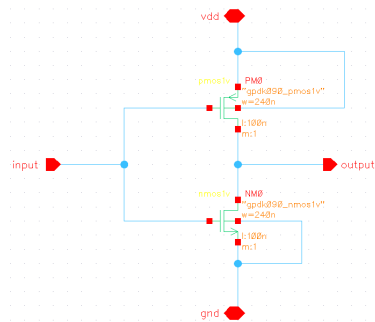


Fig. 4. Schematic diagram of INVERTER.

Fig. 5. Gate level schematic design of 4 input or gate

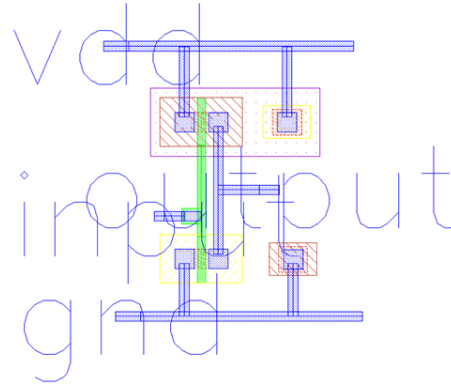


Fig. 6. Layout of INVERTER.

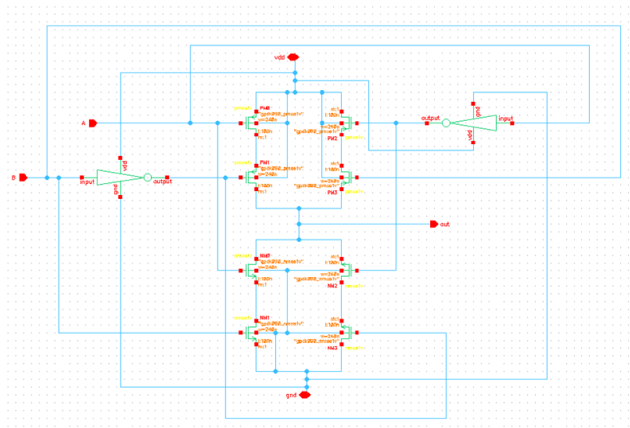


Fig. 7. Schematic diagram of XOR gate.

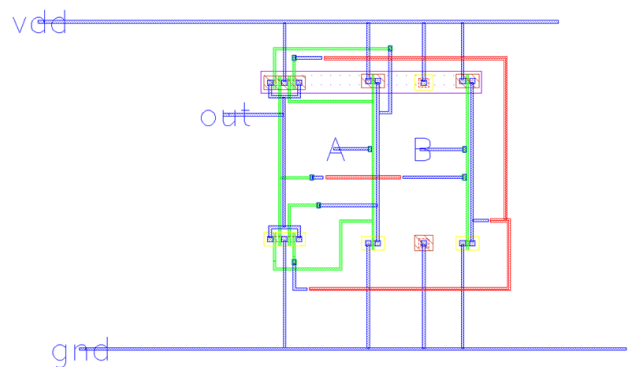


Fig. 8. Layout of XOR gate

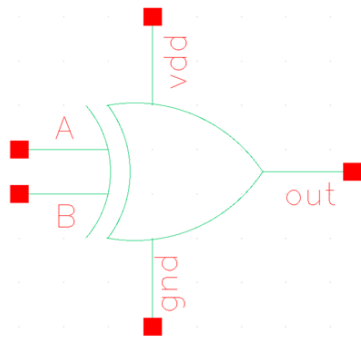


Fig. 9. Symbol of XOR gate.

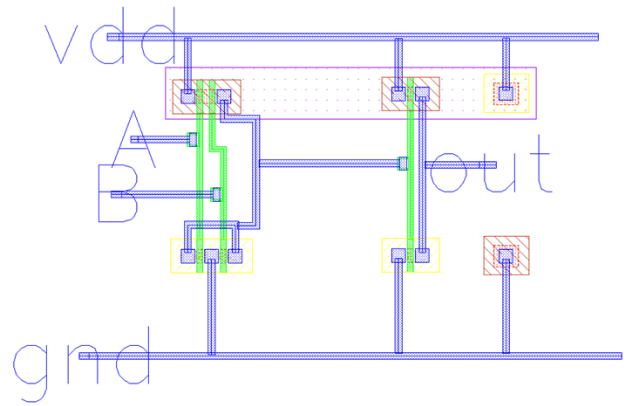


Fig. 12. Layout of OR gate.

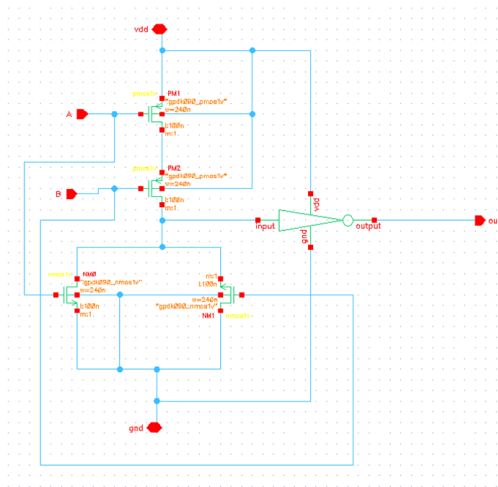


Fig. 10. Schematic diagram of OR gate.

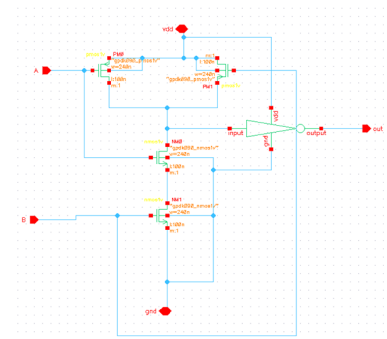


Fig. 13. Schematic of AND gate

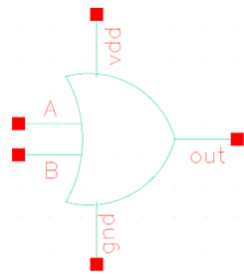


Fig. 11. Symbol of OR gate.

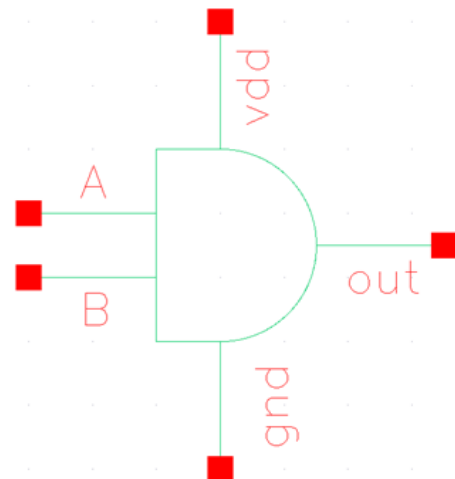


Fig. 14. Symbol of AND gate.

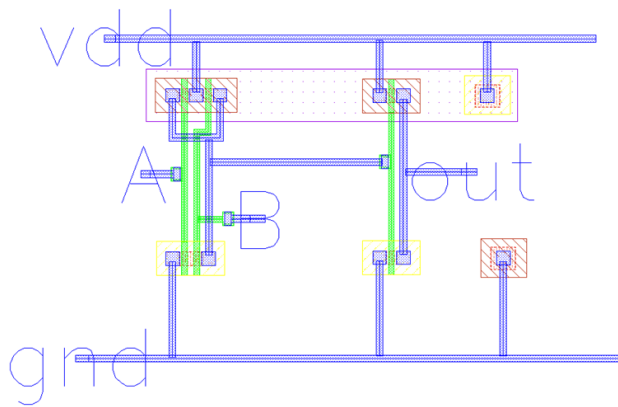


Fig. 15. Layout of AND gate.

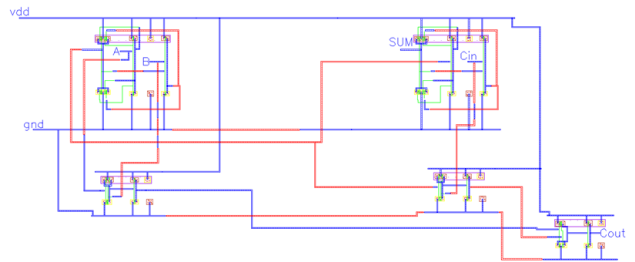


Fig. 18. Layout of 1-bit Full Adder.

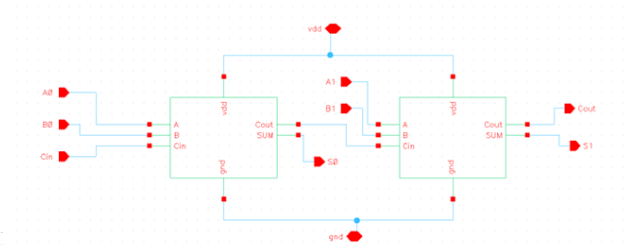


Fig. 19. Schematic of 2-bit Full Adder.

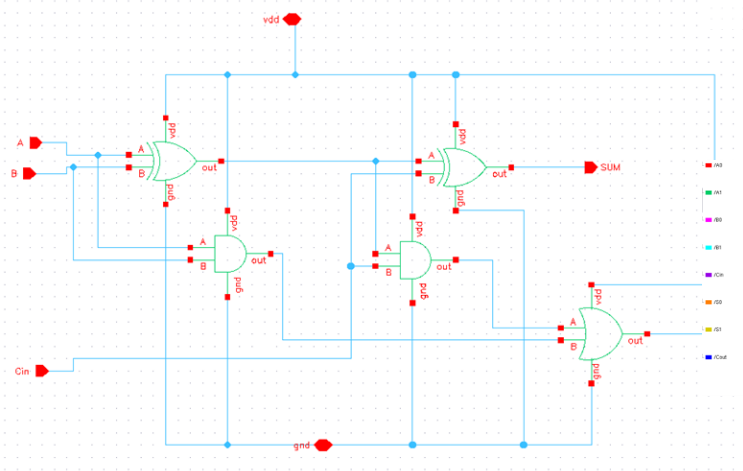


Fig. 16. Schematic of 1-bit Full Adder

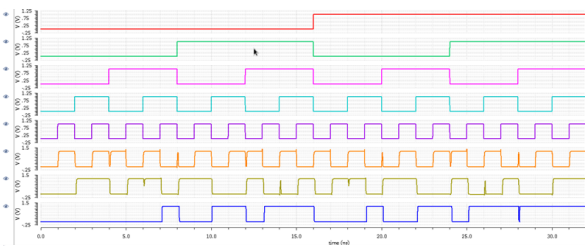


Fig. 20. Output of 2-bit Full Adder (Schematic)

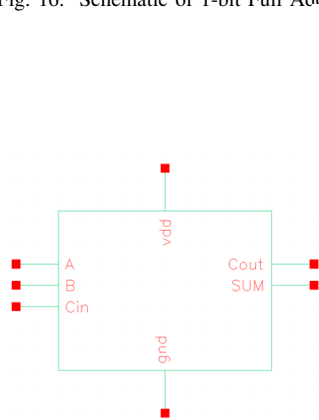


Fig. 17. Symbol of 1-bit Full Adder.

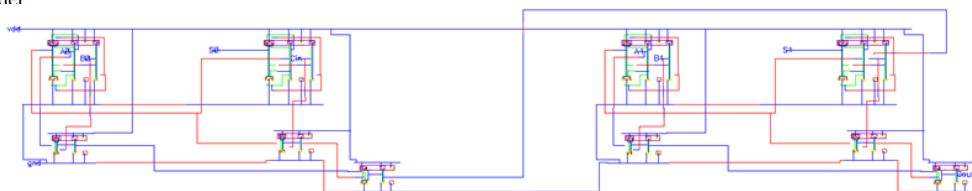


Fig. 21. Layout of 2-bit Full Adder.



Fig. 22. DRC Result of 2-bit Full Adder.

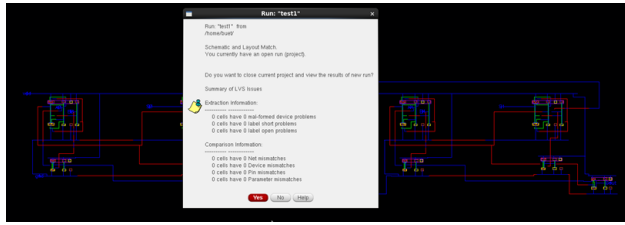


Fig. 23. LVS Result of 2-bit Full Adder(LVS Check)

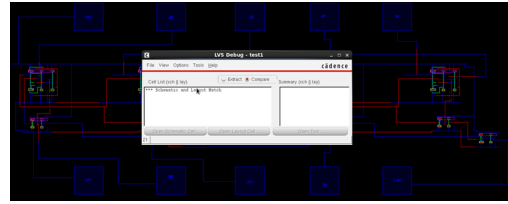


Fig. 28. LVS Result of 2-bit Full Adder after adding I/O Pad.

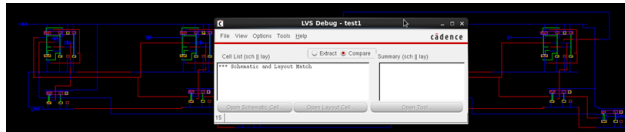


Fig. 24. LVS Result of 2-bit Full Adder(Schematic and Layout Match)

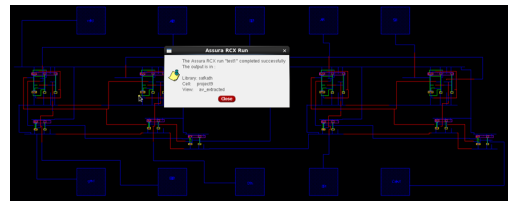


Fig. 29. RCX Result of 2-bit Full Adder after adding I/O Pad.

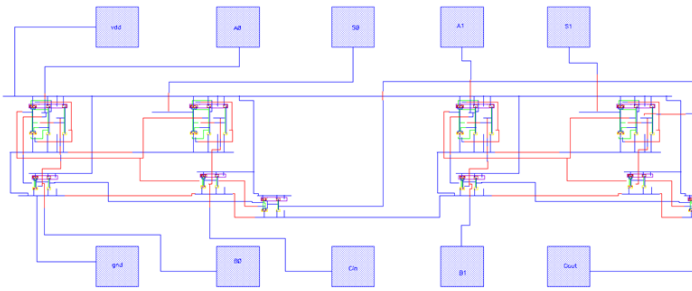


Fig. 25. Layout of 2-bit Full Adder after adding I/O Pad.

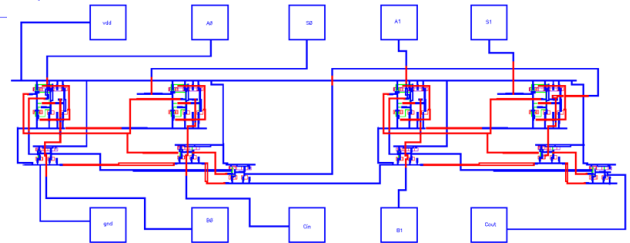


Fig. 30. AV Extracted Layout of 2-bit Full Adder after adding I/O Pad.

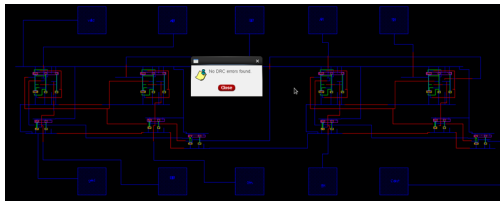


Fig. 26. DRC Result of 2-bit Full Adder after adding I/O Pad.

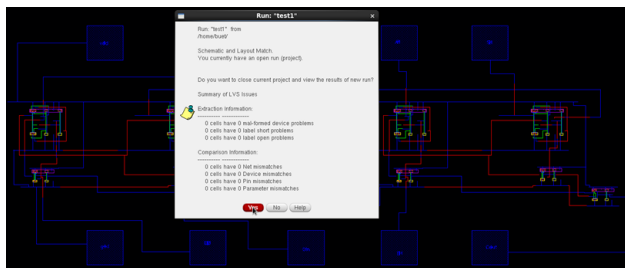


Fig. 27. LVS before adding Padding

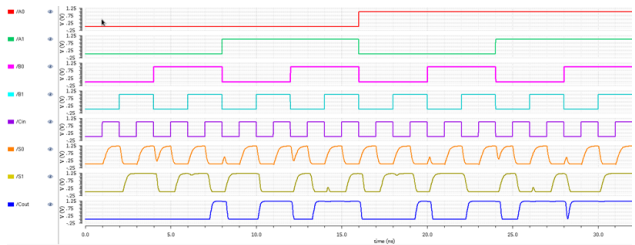


Fig. 31. Post Layout Simulation Output of 2-bit Full Adder

III. RESULT ANALYSIS & DISCUSSION

The design and implementation of the 2-bit full adder project provided a comprehensive hands-on experience in digital integrated circuit (IC) design, encompassing every stage from system-level conceptualization to post-layout validation using Cadence Virtuoso and the GPDK 90nm CMOS technology. The full adder, a fundamental building block of arithmetic and logic units (ALUs), serves as a critical element in more complex systems such as processors, multipliers, and digital counters. By constructing a 2-bit version, the project extends the learning from a single full adder to a hierarchical, modular design process, aligning with modern VLSI methodology.

The project began with the functional analysis of the full adder logic, identifying the relationship between inputs (A1, A0, B1, B0, Cin) and outputs (S1, S0, Cout). This was followed by the derivation of Boolean expressions, which were then implemented using gate-level and subsequently transistor-level schematics. Each component—AND, OR, and XOR gates—was carefully constructed and verified to ensure accuracy at every level of abstraction.

The architectural design employed a hierarchical approach, building two 1-bit full adders and cascading them to form the 2-bit structure. Symbol creation, modular reuse, and schematic connectivity were key to managing complexity and improving clarity. Once verified through pre-layout simulation, we advanced to layout generation.

The layout phase introduced new challenges, including DRC (Design Rule Check), LVS (Layout Versus Schematic), and parasitic extraction. Errors encountered in layout, such as unconnected nets or violations in metal spacing, were carefully resolved. Successful DRC and LVS results validated the correctness of the physical design with respect to the logical schematic.

The post-layout simulation phase introduced complexity into the system by accounting for parasitic capacitance and resistance extracted from the layout. The waveform analysis confirmed that although minor glitches and transition delays were present (due to parasitic coupling), the outputs (S0, S1, Cout) remained logically consistent with expectations for all input combinations. This demonstrated the robustness and manufacturability of the design.

Importantly, this project highlighted the differences between idealized simulation and physical implementation, a key insight in VLSI design. It showed how parasitic effects, often negligible in early design, become critical to performance and reliability in post-layout stages.

IV. APPLICATIONS

The 2-bit full adder, while simple, is a fundamental component in digital electronics. Its applications include:

- Arithmetic Logic Units (ALUs)
- Binary Counters
- Multipliers
- Digital Signal Processing (DSP)
- Control Units
- Custom Hardware (ASIC/FPGA)

V. CONCLUSION

The design and simulation of the 2-bit full adder were successfully completed using Cadence Virtuoso and the GPDK 90nm technology. The project involved the creation of a modular structure using two 1-bit full adders and verified the logical correctness of the circuit through both schematic-level and post-layout simulations. Design Rule Checks (DRC) and Layout Versus Schematic (LVS) checks were passed without errors, confirming layout integrity. Post-layout simulation showed expected outputs, with minor delay effects due to parasitics. The project demonstrated the complete digital IC design flow and provided practical experience in working with a standard CMOS process, layout tools, and signal analysis.

VI. ACKNOWLEDGEMENTS

All praise and gratitude are due to the Almighty for granting us the strength, patience, and opportunity to complete this project successfully. We would like to express our sincere appreciation to our respected teacher **Mr. Muhammad Mahmudul Hasan Tareq**, Assistant Professor, Department of Electrical and Electronic Engineering, CUET, for his continuous guidance, encouragement, and insightful suggestions throughout the course of this work. We are also deeply thankful to **Dr. Rabiul Islam**, Assistant Professor, Department of Electrical and Electronic Engineering, CUET, for his valuable support and direction during the execution of this project. Their mentorship and feedback played a vital role in helping us achieve our objectives and improve our understanding of digital circuit design.